Intermediate status report: House Prices – Advanced Regression Techniques

DAM2026 · checkpoint 13 November 2026 · Majba ull Alam Akash, Kofi Opoku Ayim, Omisore Omolola Bimpe

Competition: kaggle.com/competitions/house-prices-advanced-regression-techniques. Predict the sale price of houses in Ames, Iowa from 79 features (size, quality, location, age, ...).

Metric: RMSE between the logarithms of predicted and actual prices, so errors are measured relative to the price. All models are trained on log(1 + SalePrice).

Repository: github.com/majbaulalamhse-code/Kaggle-Competition-House_price_Prediction

1. What has been done

Data. 1,460 training and 1,459 test houses, 79 features (36 numeric, 43 categorical). Validation: 5-fold cross-validation (KFold, shuffle, seed 42) with the competition metric.

Key EDA findings (notebook 01, Kofi)

- The price is right-skewed (median $163,000, max $755,000, skewness 1.88); after the log transform it is almost symmetric (0.12).
- Quality and size drive the price: correlation with log price 0.82 for OverallQual, 0.70 for living area, 0.68 for garage size; price grows faster than linearly with quality. Location matters too (about 3× between neighbourhoods).
- 34 columns have missing values, mostly meaning "the house does not have it" (PoolQC 99.5%, MiscFeature 96%, Alley 94%, Fence 81%), filled with "None"/0; LotFrontage (18%) is filled with the neighbourhood median. 15 columns have missing values only in test.
- Two huge but cheap houses (4,676 and 5,642 sq ft, partial sales) are outliers and are removed from training; Utilities is almost constant and dropped.

Preprocessing and models (House_Price_2.ipynb, Akash). Quality columns are encoded as ordinal numbers, other categories one-hot; skewed numeric columns are Box-Cox transformed and scaled with RobustScaler for the linear and kernel models. 14 new features (total area, bathrooms, age, flags, interactions). Lasso, ElasticNet, SVR and KernelRidge are tuned by grid search on CV; eight model types are combined in a stacking blend. Based on the public notebook "Stacked Regressions" by Serigne (referenced in the README).

Leakage-free validation (notebook 03, Omisore). All preprocessing re-fitted inside each fold: the linear models change by only 0.0003, so the CV results are reliable. A fragile Box-Cox rule left PoolArea untransformed in two folds and broke SVR for houses with a pool (0.1217); with the fixed rule SVR reaches 0.1103. Total-area features are the most useful engineered features; the interaction features do not help.

2. Current scores

model (5-fold CV, RMSE on log price)	CV RMSE
Constant (mean log price): trivial baseline	0.3988
Rule: median price by OverallQual	0.2305
Ridge	0.1134
Gradient boosting	0.1126
Lasso	0.1120
XGBoost	0.1120
SVR (best single model)	0.1108
Stacking blend (Lasso, SVR, GBR, XGBoost)	**0.1063**

Kaggle submission	CV RMSE	Kaggle score
submission.csv (weighted ensemble)	0.1080	0.12232
submission_v2.csv (stacking blend)	**0.1063**	**0.12200**

The Kaggle score is about 0.016 higher than the CV score: the blend weights were chosen on the same training data, and the test set may contain unusual houses like the removed outliers. The ranking of the two submissions is the same on CV and Kaggle.

3. Plan until the defense (4 and 11 December)

- Choose the blend weights inside the cross-validation (nested) to remove the optimism of the CV score and narrow the gap to Kaggle.
- Tune the tree models (gradient boosting, XGBoost, LightGBM) with random search on the leakage-free CV.
- Try neighbourhood-level features (e.g. median price per neighbourhood computed inside the folds) and remove features that do not help.
- Prepare the 15–20 min presentation; every member presents their own part.

member	done so far	until the defense
Majba ull Alam Akash	preprocessing, features, tuning, stacking, Kaggle submissions (House_Price_2.ipynb)	tuning of the tree models, final blend and submission; slides on models and results
Kofi Opoku Ayim	data description, EDA, missing values, outliers, baseline (01_eda_baseline.ipynb); README	neighbourhood-level features and their EDA; slides on problem, data and EDA
Omisore Omolola Bimpe	leakage-free validation, feature effect, model comparison (03_validation_features.ipynb)	nested blend weights, validation vs Kaggle analysis; slides on validation and comparison

4. Key figures

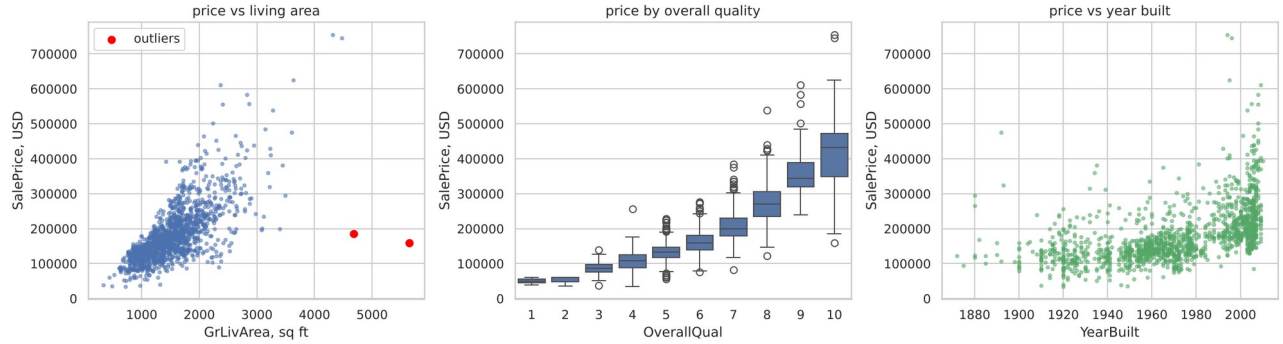


Figure 1. Price vs living area (red: the two removed outliers), by overall quality, and vs year built (notebook 01).

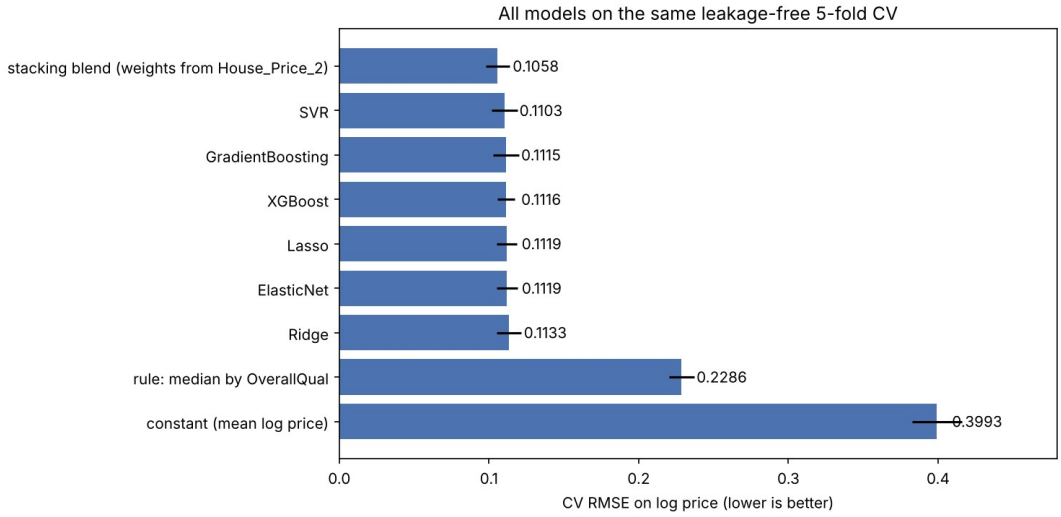


Figure 2. All models on the same leakage-free 5-fold CV (notebook 03); error bars = std over folds.